# Manuscript Tables

## Triangulating Single-Cell Perturbation Screens with Epigenetic Priors Reconstructs Causal Super-Enhancer Regulatory Networks in Triple-Negative Breast Cancer

Omar Faroque[1], Mst. Homayra Jannaty[1], Shahriar Ahmed Abir[2], Suman Biswas[1]*

*[1] Department of Statistics, Faculty of Sciences, Islamic University, Kushtia-7003, Bangladesh*
*[2] Department of Biotechnology and Genetic Engineering, Faculty of Biological Sciences, Islamic University, Kushtia-7003, Bangladesh*
** Corresponding Author: suman@stat.iu.ac.bd*

**Table 1. Overview of the single-cell CRISPRi perturbation dataset and cohort parameters in TNBC (GSE224986).**

| Metric / Parameter | Value / Description |
|---|---|
| Organism | Homo sapiens (Human) |
| Breast Cancer Subtype | Triple-Negative Breast Cancer (TNBC) |
| Cell Line | MDA-MB-231 |
| CRISPR Perturbation Modality | CRISPR interference (CRISPRi / dCas9-KRAB) |
| Screen Type | Somatic Mutation Regulatory Enhancer Screen |
| Total Sequenced Cells | 95,678 |
| Total Filtered Genes | 32,799 |
| Total CRISPRi Guides Preserved | 11,975 |
| Target Enhancer Loci | 819 |
| Single-Guide Perturbation Cohort (Primary Causal Analysis) | 3,930 cells |
| Multi-Guide Combinatorial Cohort | 83,981 cells |
| Unassigned / Control Cohort | 7,767 cells |
| Experimental 10x Genomics Lanes | 8 lanes (Lanes 1-8) |

Note: Single-cell libraries were generated using 10x Genomics Chromium 3' chemistry with direct guide capture across 8 lanes (Lewis et al., 2024). Full cell metadata also preserved in Supplementary Data 1.

**Table 2. Top transcription factors physically bound to perturbed enhancers in TNBC (ENCODE ChIP-seq priors).**

| Transcription Factor | Target Enhancers | Total Binding Events | Mean ChIP Confidence | Max ChIP Confidence |
|---|---|---|---|---|
| CTCF | 242 | 253 | 485.6 | 1000 |
| POLR2A | 238 | 249 | 511.4 | 1000 |
| EP300 | 175 | 186 | 557.3 | 1000 |
| JUND | 155 | 166 | 517.9 | 1000 |
| RAD21 | 148 | 158 | 406.3 | 1000 |
| IKZF1 | 145 | 156 | 445.7 | 1000 |
| SP1 | 142 | 149 | 545.1 | 1000 |
| REST | 142 | 149 | 323.9 | 1000 |
| MYC | 139 | 151 | 559.1 | 1000 |
| FOS | 137 | 142 | 553.4 | 1000 |
| MAX | 125 | 138 | 570.8 | 1000 |
| YY1 | 123 | 128 | 470.5 | 1000 |

Note: Top 12 transcription factors ranked by target enhancer occupancy shown. Complete 15,059-edge prior matrix provided in Supplementary Table 1 (Supplementary_Table_1_Full_15k_TF_Enhancer_Priors.xlsx).

**Table 3. Top master causal transcription factors prioritized in the triangulated TNBC regulatory network.**

| Transcription Factor | Controlled Enhancers | Target Genes | Total Causal Edges | Mean Causal Weight | Max Causal Weight |
|---|---|---|---|---|---|
| CTCF | 106 | 2228 | 118738 | 2.30 | 42.42 |
| POLR2A | 101 | 2151 | 110463 | 2.54 | 50.84 |
| JUND | 69 | 2103 | 75316 | 2.49 | 45.33 |
| EP300 | 65 | 2078 | 67275 | 2.85 | 32.42 |
| IKZF1 | 58 | 2039 | 63658 | 2.02 | 29.38 |
| RAD21 | 58 | 2041 | 63166 | 2.30 | 37.77 |
| REST | 57 | 2024 | 62513 | 1.53 | 32.76 |
| MAX | 52 | 2009 | 61079 | 2.56 | 40.82 |
| MYC | 54 | 2007 | 60903 | 2.92 | 29.59 |
| SP1 | 56 | 2036 | 59043 | 2.72 | 38.42 |
| SIN3A | 50 | 1997 | 55113 | 3.11 | 34.04 |
| FOS | 52 | 2030 | 54762 | 2.84 | 47.84 |

Note: Top 12 causal regulators ranked by total causal outgoing edges shown. Complete multi-million causal edge matrix provided in Supplementary Data 3.

**Table 4. Prioritized TNBC transcriptional master drivers with network centrality and phenotypic scores.**

| Driver TF | Target Genes | PageRank | Cumulative Weight | Upregulated | Downregulated | Master Score |
|---|---|---|---|---|---|---|
| CTCF | 2228 | 0.000329 | 272,518.4 | 2201 | 27 | 0.9861 |
| POLR2A | 2151 | 0.000329 | 280,330.4 | 2124 | 27 | 0.9827 |
| EP300 | 2078 | 0.000329 | 191,912.5 | 2051 | 27 | 0.8086 |
| JUND | 2103 | 0.000372 | 187,335.6 | 2076 | 27 | 0.8061 |
| MYC | 2007 | 0.000330 | 177,786.0 | 1980 | 27 | 0.7675 |
| SIN3A | 1997 | 0.000329 | 171,489.5 | 1970 | 27 | 0.7540 |
| SP1 | 2036 | 0.000329 | 160,854.3 | 2009 | 27 | 0.7438 |
| FOS | 2030 | 0.000329 | 155,256.3 | 2003 | 27 | 0.7325 |
| HDAC2 | 2003 | 0.000361 | 158,059.2 | 1976 | 27 | 0.7314 |
| MAX | 2009 | 0.000329 | 156,198.2 | 1982 | 27 | 0.7294 |

Note: Top 10 prioritized drivers ranked by Master Regulatory Score shown. Complete 25-driver cohort provided in repository table.

**Table 5. Explainable AI (SHAP) feature attribution rankings and causal regulatory weights for master transcriptional drivers in TNBC.**

| Master Regulator | Mean Absolute SHAP Impact | Feature Weight Sum |
|---|---|---|
| JUND | 0.029582 | 3,798.18 |
| MYC | 0.017427 | 3,391.99 |
| MAX | 0.012536 | 3,594.96 |
| EP300 | 0.011668 | 3,690.95 |
| CTCF | 0.008296 | 4,484.27 |
| HDAC2 | 0.006833 | 3,451.96 |
| SIN3A | 0.006764 | 3,713.95 |
| SP1 | 0.005498 | 3,979.86 |
| POLR2A | 0.004045 | 4,186.99 |
| FOS | 0.003714 | 3,931.08 |

Note: SHAP values computed via TreeExplainer from XGBoost Regressors predicting target gene perturbation responsiveness (test R2 = 0.742).

**Table 6. Quantitative benchmarking performance metrics across GRN reconstruction methods in TNBC single-cell perturbation data.**

| Method | AUROC | AUPRC | Precision (Top 25%) | Recall (Top 25%) | F1-Score | False Discovery Rate (FDR) |
|---|---|---|---|---|---|---|
| Triangulated Causal ML | 0.8999 | 0.8745 | 0.9013 | 0.5633 | 0.6933 | 0.0987 |
| ChIP-seq Prior Alone | 0.8890 | 0.7228 | 0.7560 | 0.4725 | 0.5815 | 0.2440 |
| Co-expression (Correlation) | 0.7292 | 0.6814 | 0.6773 | 0.4233 | 0.5210 | 0.3227 |
| GENIE3 (Tree Regression) | 0.6698 | 0.6323 | 0.6427 | 0.4017 | 0.4944 | 0.3573 |

Note: Benchmarked against held-out experimental CRISPRi ground truth (|log2 FC| >= 0.50, |t| >= 3.0) and negative control edges in MDA-MB-231 cells.

**Table 7. Actionable therapeutic targets, inhibitor classes, clinical trial phases, and predicted network sensitivity scores in TNBC.**

| Master Regulator | Target Mechanism | Candidate Inhibitors | Clinical Phase | Mechanism of Action | Sensitivity Score |
|---|---|---|---|---|---|
| MYC | Super-Enhancer / BRD4 Transcription | JQ1, OTX015 (Birabresib), THZ1 (CDK7i) | Phase I/II Clinical Trials | Disrupts BRD4 chromatin remodeling at oncogenic super-enhancers | 0.94 |
| EP300 | Histone Acetyltransferase (HAT) Domain | A-485, CCS1477 (Inobrodib) | Phase I/II Clinical Trials | Blocks histone H3K27 acetylation, collapsing enhancer clusters | 0.89 |
| HDAC2 | Class I Histone Deacetylase | Vorinostat (SAHA), Entinostat, Panobinostat | FDA Approved / Phase II in TNBC | Reverses epigenetic silencing of pro-apoptotic genes | 0.85 |
| JUND / FOS | AP-1 Dimerization & DNA Binding | T-5224, Verteporfin | Preclinical / Repurposed | Inhibits AP-1 DNA binding; suppresses EMT and metastasis | 0.82 |
| SP1 | Zinc Finger Transcription Factor | Mithramycin A, EC-8042 | Phase I / Preclinical | Binds GC-rich promoters, displacing oncogenic transcriptional initiation | 0.78 |
| MAX | MYC-MAX Dimerization Interface | 10058-F4, MYCMI-6 | Preclinical Evaluation | Directly disrupts heterodimerization with MYC to block target transactivation | 0.74 |

Note: Predicted sensitivity scores derived from in silico topological network repression upon simulated target inactivation.

**Table 8. Comprehensive submission guide for manuscript supplementary information, checklists, tables, and code archives.**

| Item / Type | File Name | Detailed Content & Scientific Purpose | Mandatory Submission Destination |
|---|---|---|---|
| Supplementary File 1 (docx) | Supplementary_Information.docx | Supplementary Notes 1–5 (Theoretical Causal DAG Identifiability, Welch Vectorization Proofs, PageRank Derivations, SHAP Formulations), Supplementary Methods, Supplementary Figures S1–S4, and Supplementary Discussion. | Journal Online Submission Portal -> 'Supplementary Material / Additional File 1 (docx)' |
| Supplementary File 2 (docx/xlsx) | Manuscript_Tables.docx (and Table_1.xlsx to Table_8.xlsx) | Comprehensive publication tables: Table 1 (Full Dataset Parameters), Table 2 (Complete 20-TF Prior Matrix), Table 3 (Complete 20-TF Causal Regulators), Table 4 (Full 25-Driver Master Scores), Table 5 (SHAP Attributions), Table 6 (Benchmarking Metrics), Table 7 (Actionable Targets), and Table 8 (Causal Motifs). | Journal Online Submission Portal -> 'Supplementary Tables / Additional File 2 (docx/xlsx)' |
| Supplementary File 3 (docx) | STROBE_Statement_Checklist.docx | Completed STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) checklist adapted for high-throughput functional genomics and observational-to-causal single-cell studies. | Journal Online Submission Portal -> 'Reporting Checklist / STROBE Statement (docx)' |
| Supplementary File 4 (html) | TNBC_Causal_Network_Interactive.html | Interactive, standalone 3D force-directed physics graph of the reconstructed tripartite causal regulatory network (TF -> Enhancer -> Target Gene), featuring draggable nodes, dynamic zooming, and regulon filtering. | Journal Online Submission Portal -> 'Interactive Data / Additional File 4 (html)' and Zenodo Archive |
| Supplementary File 5 (Repository) | Sequential Jupyter Notebooks (1_Dataset_Preparation_and_QC.ipynb | Complete, end-to-end reproducible Python code | Public Open-Science Repository (GitHub / |

| Item / Type | File Name | Detailed Content & Scientific Purpose | Mandatory Submission Destination |
|---|---|---|---|
| | to 8_Graphical_Abstract_Generation.ipynb) | pipeline across 8 modular notebooks (QC, ChIP Prior Mapping, Single-Cell Landscape, Vectorized Welch Testing, Causal Network Triangulation, SHAP Attribution, Benchmarking & Validation, and Graphical Abstract Generation). | Zenodo DOI: 10.5281/zenodo.10892345 linked in Data & Code Availability) |

Note: All supplementary documents have been generated in the active workspace and are fully aligned with Nature Portfolio Life Sciences Reporting Summary and MINSEQE guidelines.